Do you know how many people are diagnosed with Parkinson’s every year in the US?

Using AI: Making Parkinson's non-existent

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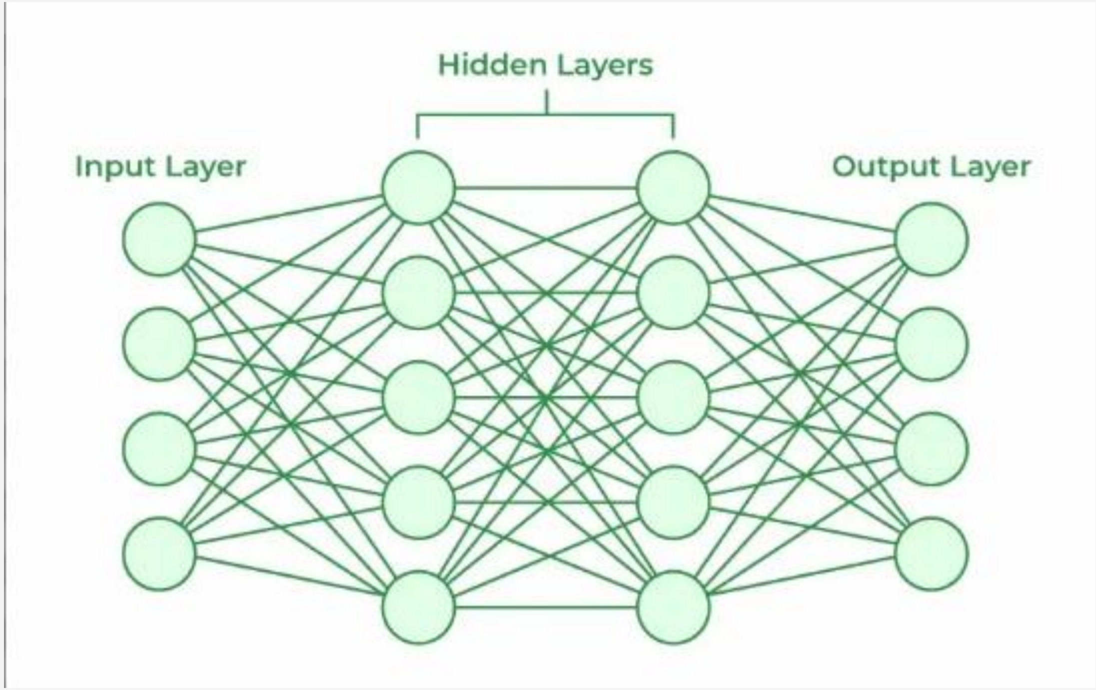
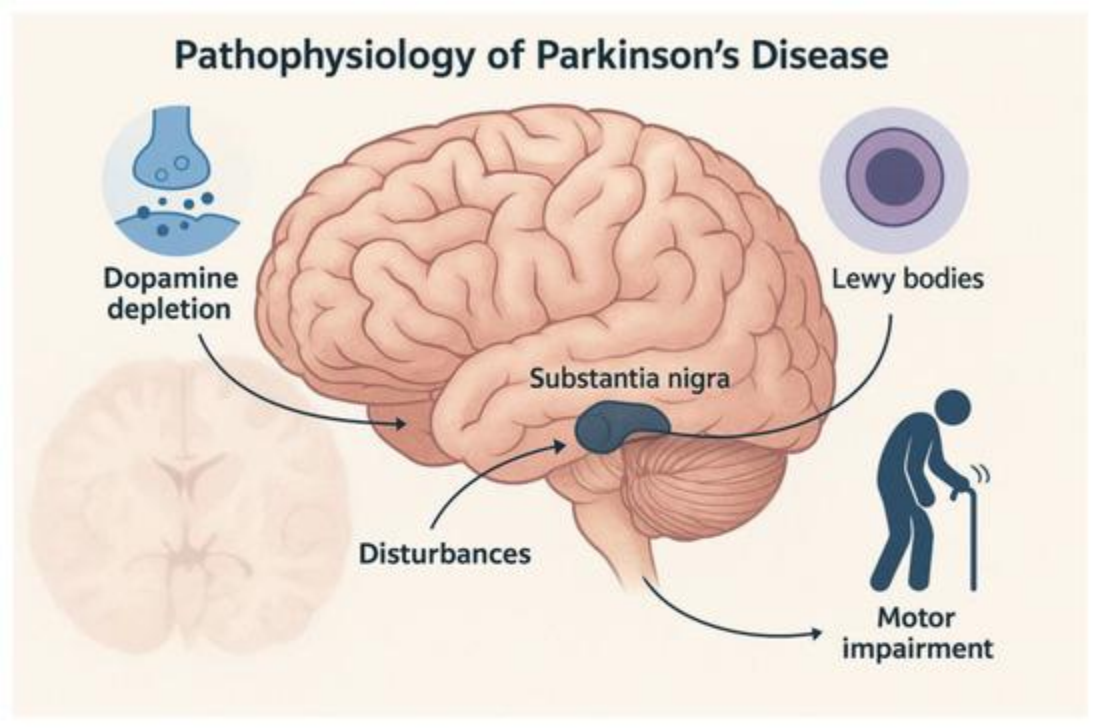
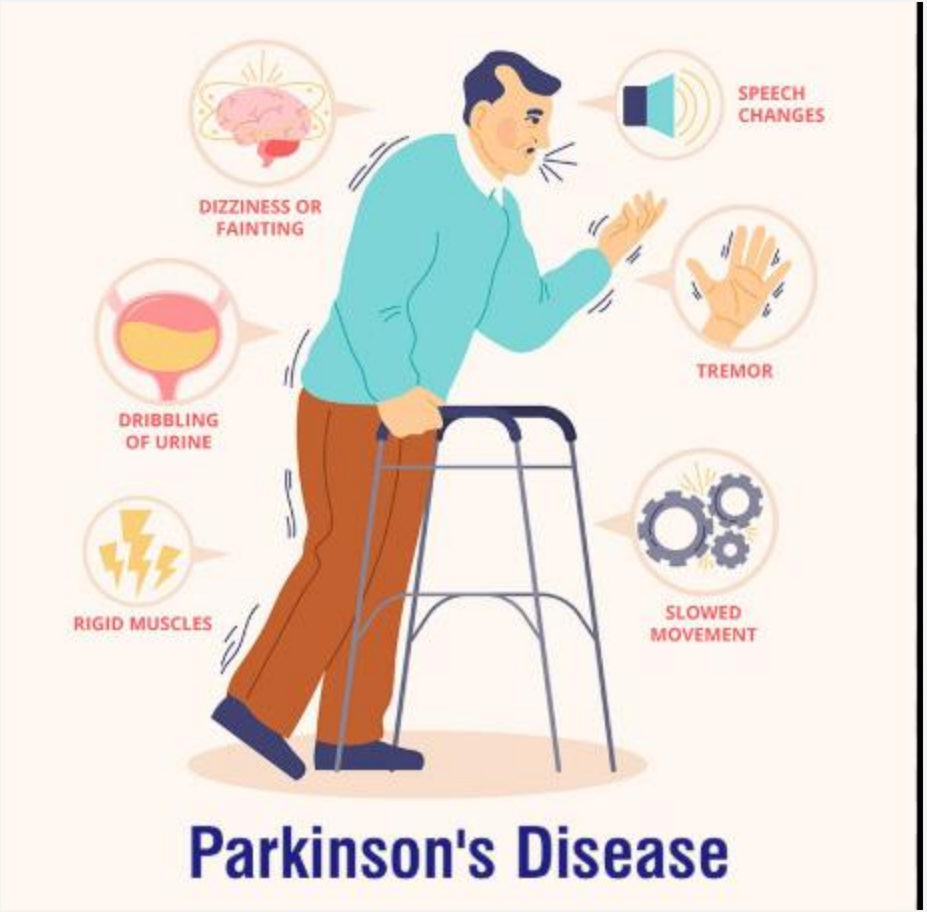
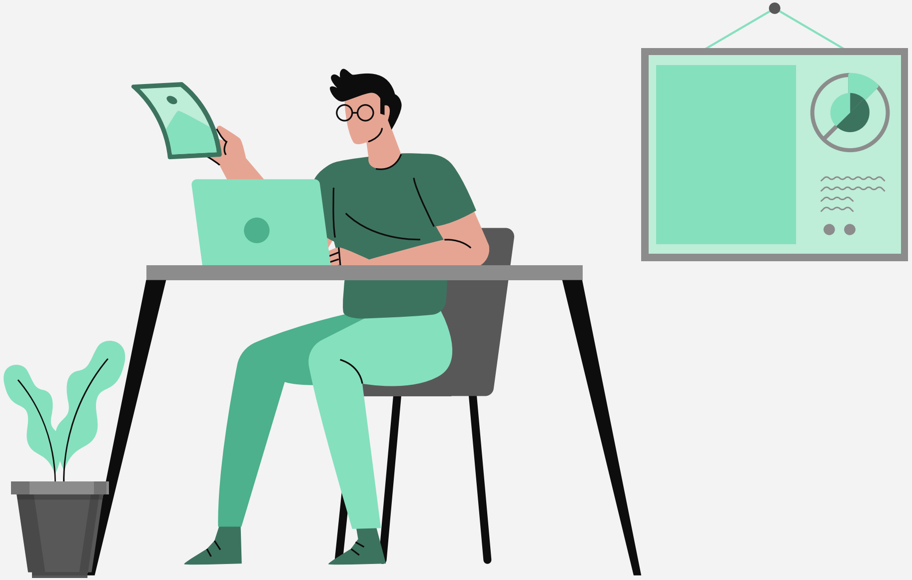
Background



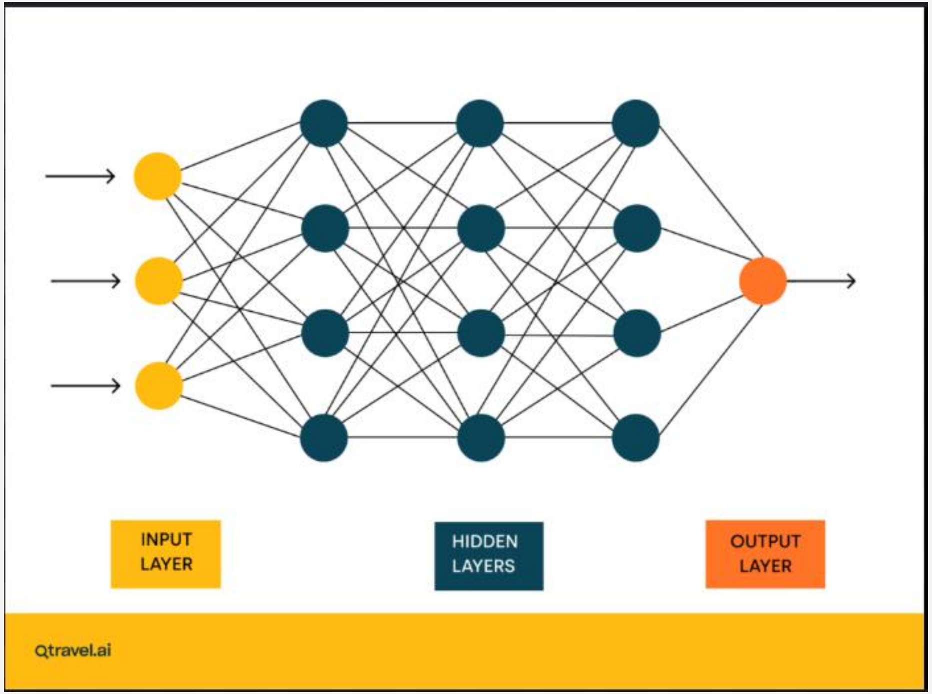

Pathophysiology of Parkinson's Disease
Dopamine depletion
Lewy bodies
Substantia nigra
Disturbances
Motor impairment


DIZZINESS OR FAINTING
SPEECH CHANGES
TREMOR
DRIBBLING OF URINE
RIGID MUSCLES
SLOWED MOVEMENT
Parkinson's Disease

The Question



How might Artificial Intelligence be used to detect and treat Parkinson's disease?

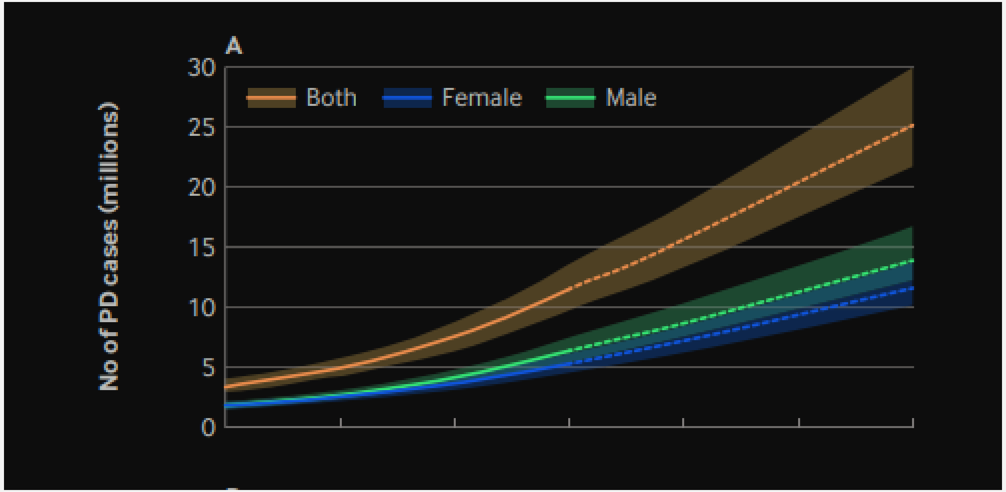


My goals

Lower the number of Parkinson's cases by 25% in the next ten years

What’s so bad about Parkinson’s?

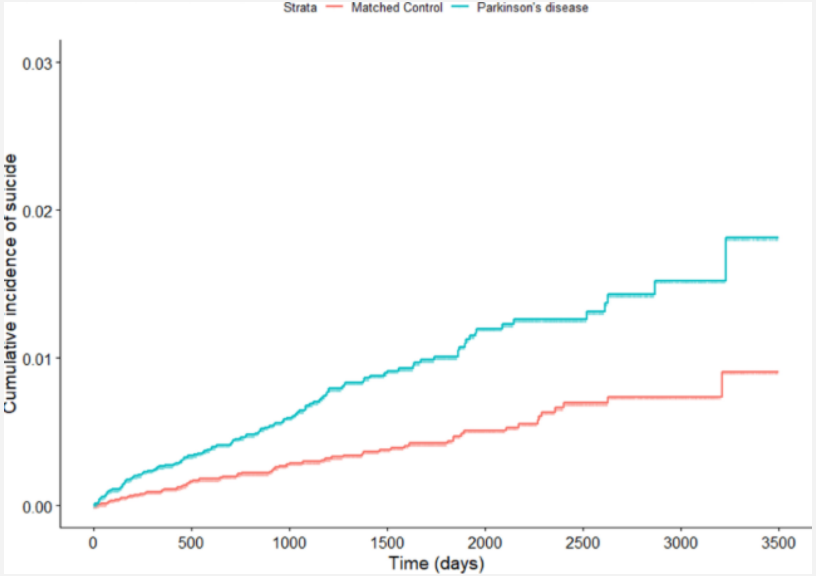
Parkinson’s is the fastest growing neurological disease

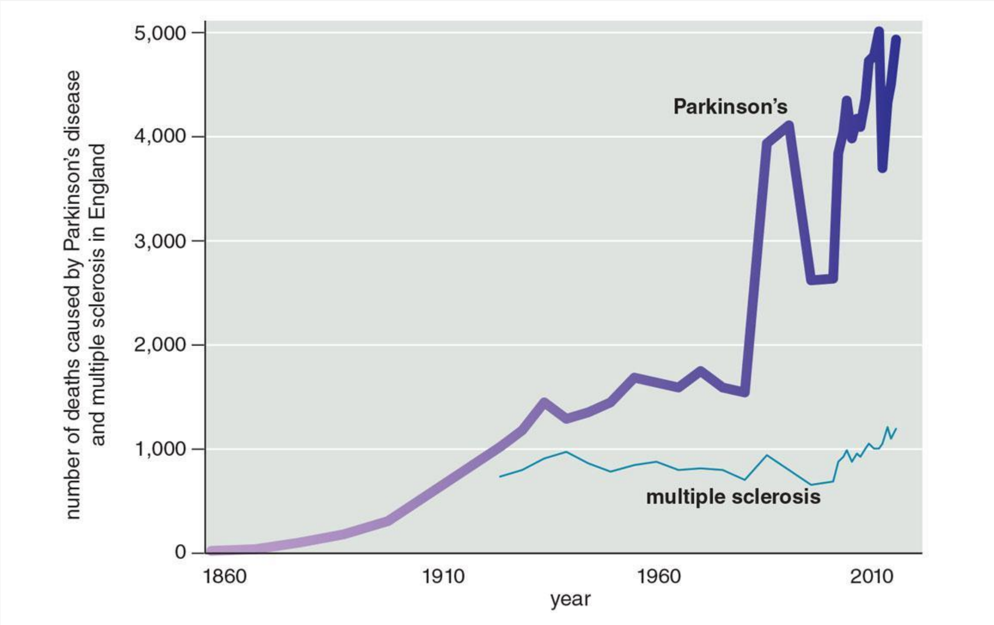


Su, Dongning, et al. "Projections for prevalence of Parkinson’s disease and its driving factors in 195 countries and territories to 2050: modelling study of Global Burden of Disease Study 2021." bmj 388 (2025).

People with Parkinson’s have a higher risk of suicide

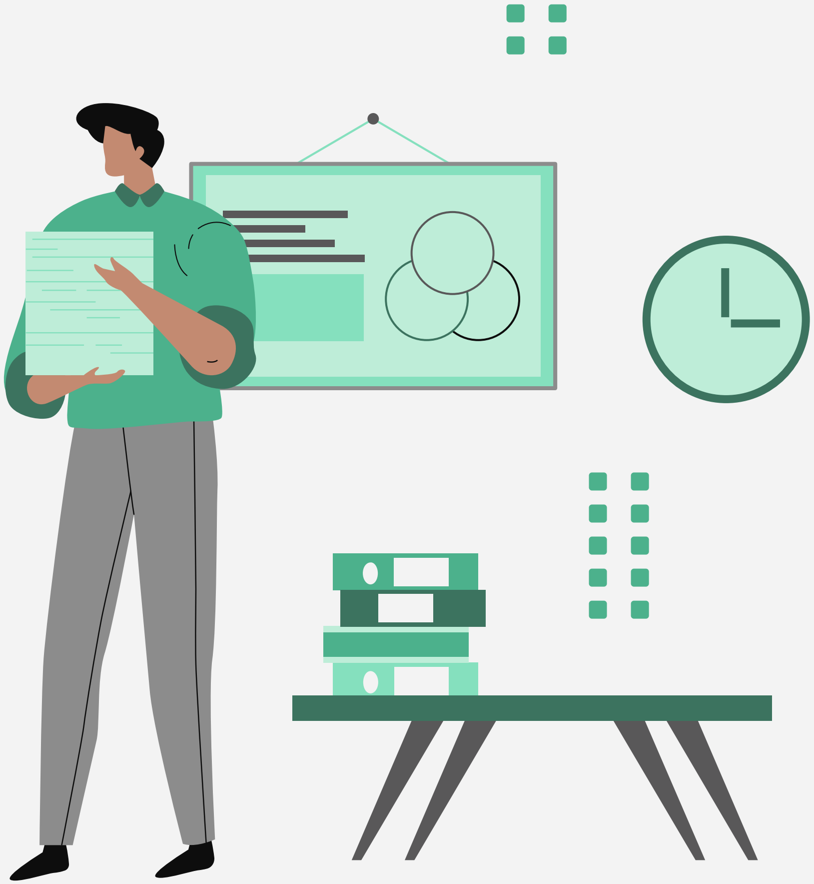
Quality of life is really bad





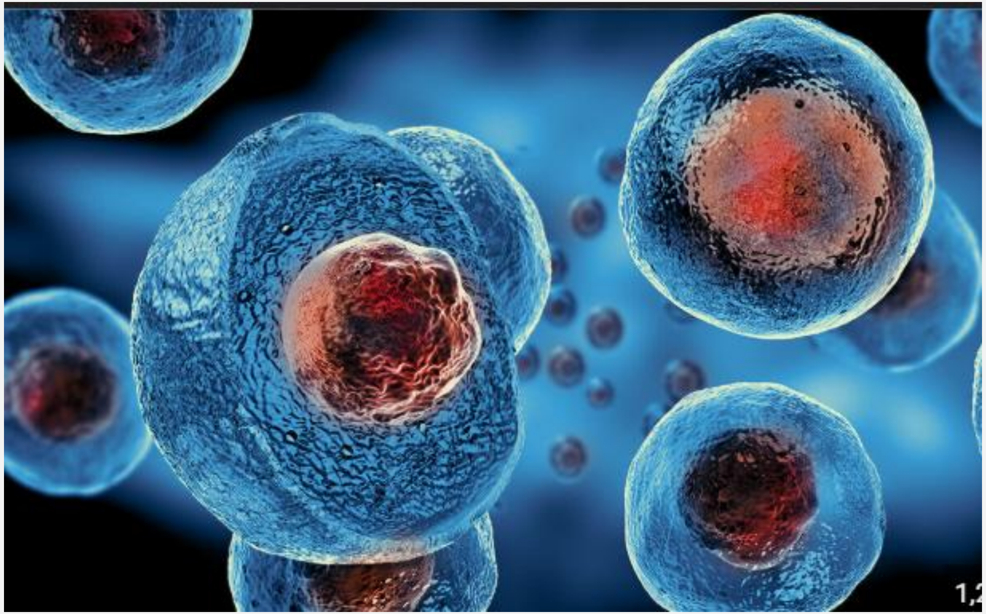
number of deaths caused by Parkinson's disease and multiple sclerosis in England
5,000
4,000
3,000
2,000
1,000
0
Parkinson's
multiple sclerosis
1860
1910
1960
2010
year

Possible Treatmen t



Stem Cell Therapy

The one with the most potential



Gene Therapy

Motion Capture

Effective Diagnosis

Ethics

- Datasets
- Potential for bias
- The Black-box problem
- Data Privacy

"The longer I had the disease and was moving around the planet, I realized people were just waiting for this ethereal cure to happen.... And I fairly quickly realized that waiting was too passive."

- Davis Phinney

How did I do this Research?

All grades: Independent Research G/T

11th and 12th grade: Intern/Mentor

Also Available To Take: AP Seminar

Look Out

Research Info Night: January 8